BANK CUSTOMER CHURN SEGMENTATION & RETENTION PRIORITIZATION

Data Analytics / Data Mining Portfolio Report

Individual Project

Python | Pandas | scikit-learn | SciPy | K-Medoids | Fuzzy Logic

Portfolio Edition - 2026

1. Executive Summary

This individual project analyzes 10,000 bank customers to identify churn patterns, build interpretable customer segments, and prioritize retention outreach. The workflow combines exploratory analysis, robust preprocessing, Genetic Algorithm feature selection, K-Medoids clustering, hierarchical benchmarking, and an explainable fuzzy decision system.

Key results

- Overall churn rate is 20.4%.
- Inactive customers churn at 26.9%, compared with 14.3% for active customers.
- Germany has the highest observed country churn rate at 32.4%.
- The Genetic Algorithm selected age, tenure, balance, products_number, active_member, and estimated_salary.
- The selected feature subset achieved cross-validated F1 of 0.584 on the fixed GA sample versus 0.563 using all source features.
- K-Medoids produced four balanced customer segments; observed churn ranges from 6.7% to 32.7% across segments.
- The final High retention-priority group has an observed churn rate of approximately 33.0%.

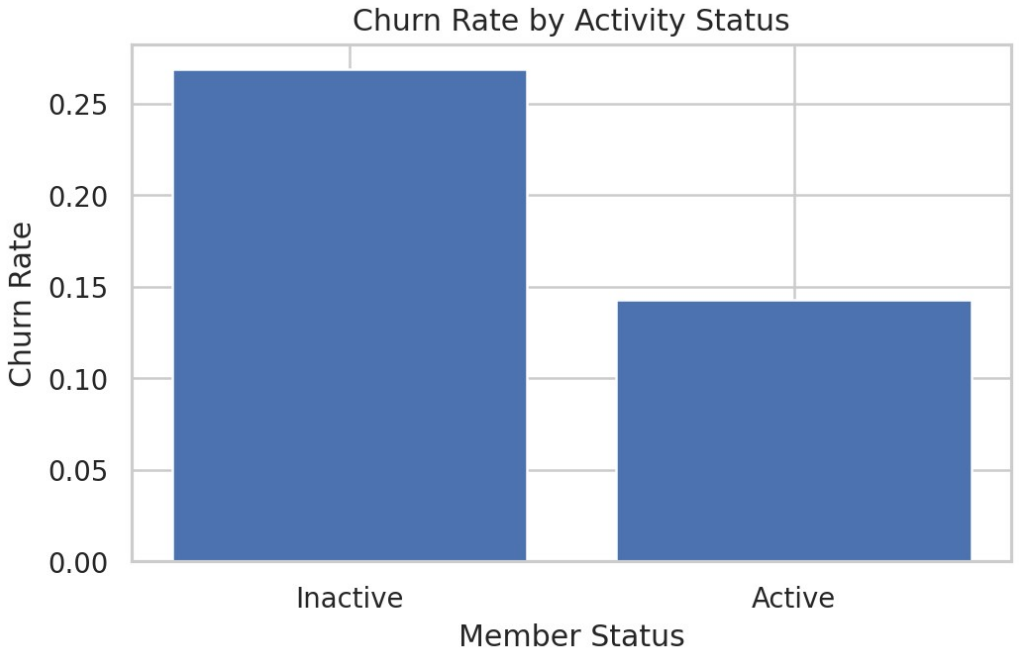


Figure 1. Customer activity is a strong actionable churn signal.

Business value

The output is designed as a decision-support workflow. Instead of treating every customer equally, retention teams can use segment profiles and priority scores to focus outreach, tailor offers, and test which interventions reduce churn most effectively.

2. Data Understanding & Exploratory Analysis

The dataset contains customer demographics, banking behavior, account balance, product usage, activity status, estimated salary, and a binary churn label. Exact duplicates are removed, missing-value fallbacks are defined, and domain checks enforce valid binary fields.

Outlier policy

Extreme values are audited using IQR boundaries but are not automatically deleted or clipped. High balances, salaries, ages, and product counts may represent legitimate customers. RobustScaler is used later to reduce the influence of extremes in distance calculations while preserving all records.

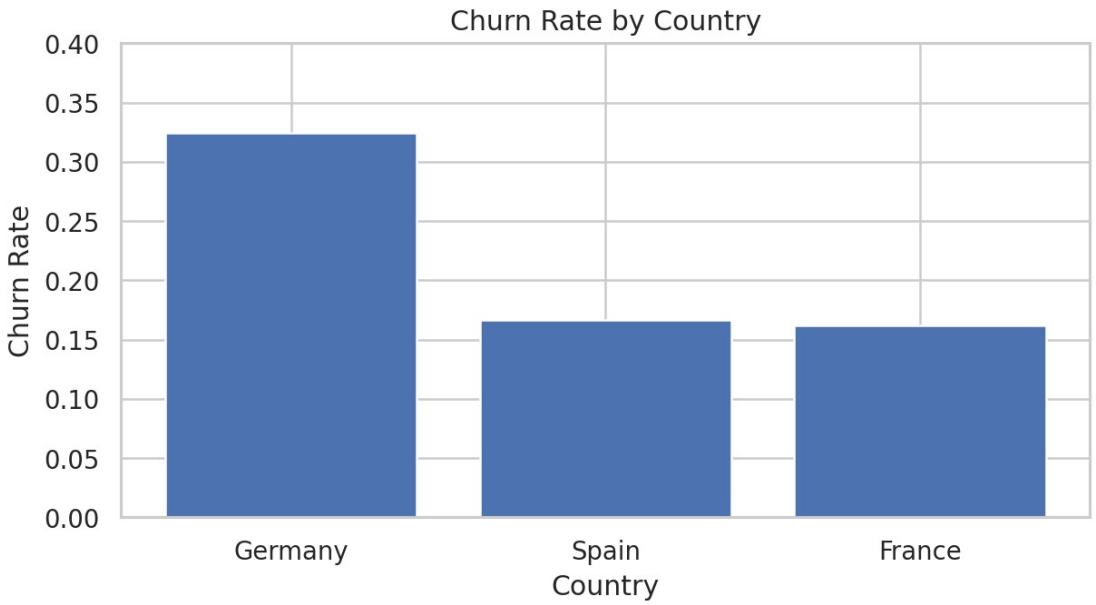


Figure 2. Observed churn rate differs materially by country.

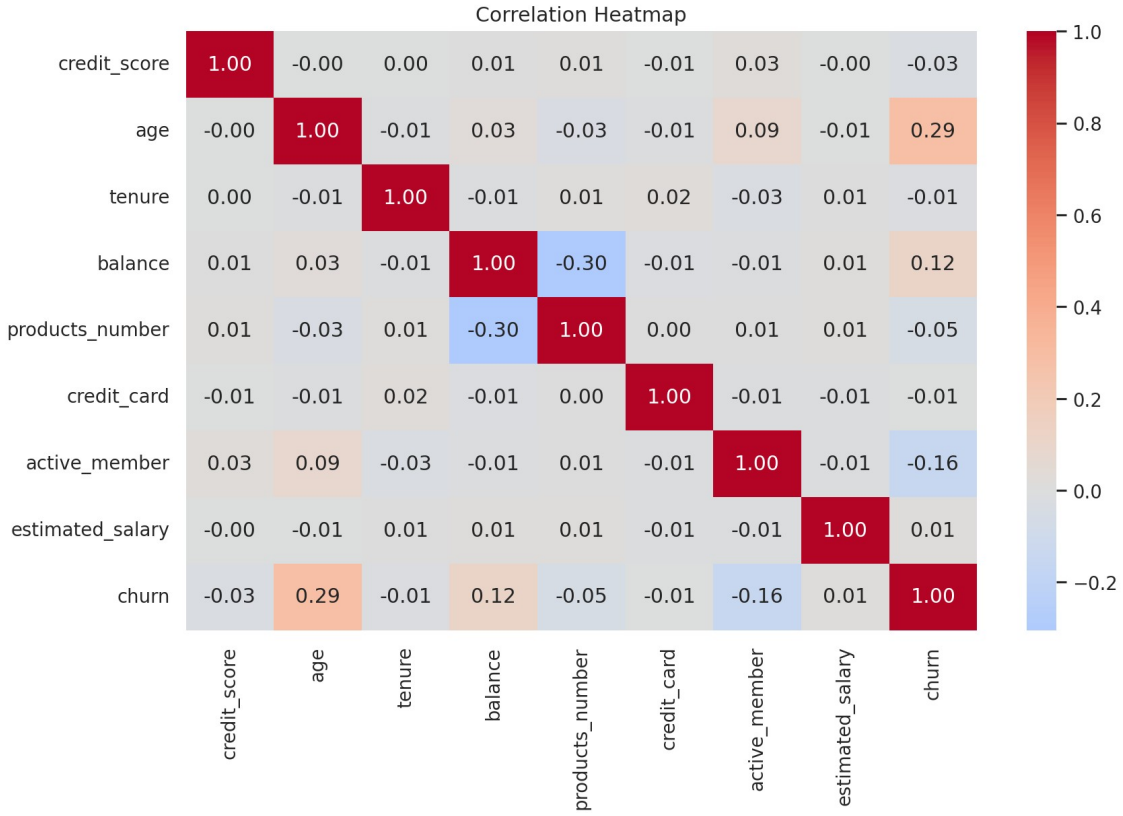


Figure 3. Numeric correlation overview used for descriptive context.

3. Preprocessing & Churn-Aware Feature Selection

Preprocessing choices

- Continuous variables are transformed with RobustScaler.
- Gender is represented as a binary feature.
- Country is one-hot encoded because country categories have no valid ordinal distance.
- Credit-card ownership and active-member status remain binary 0/1 features.
- The fitted preprocessing objects are reused for new customer records.

Genetic Algorithm

The Genetic Algorithm performs supervised, churn-aware feature selection. Its fitness signal is cross-validated F1 from a class-balanced Random Forest. This makes the later segments intentionally relevant to churn behavior. The project therefore describes the result as churn-informed segmentation rather than purely unsupervised natural grouping.

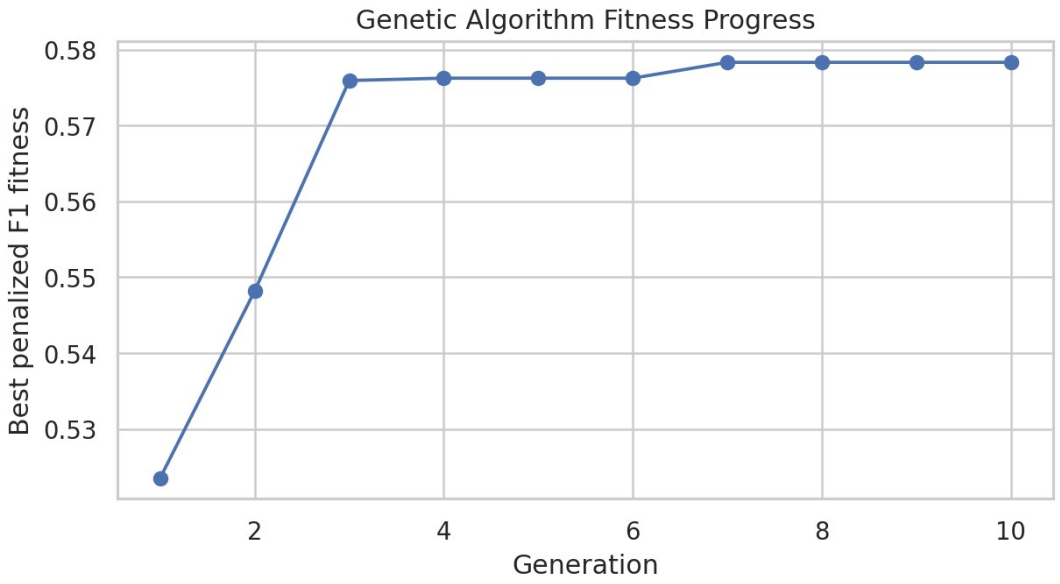


Figure 4. GA best fitness converges over generations.

Metric	Result
Selected features	age, tenure, balance, products_number, active_member, estimated_salary
Selected-feature CV F1	0.5843288821820335
All-feature CV F1	0.5627643730877562
GA evaluations	59

4. Customer Segmentation

A dependency-free alternate K-Medoids implementation is used so the project does not rely on scikit-learn-extra binary packages. Candidate cluster counts from 2 to 6 are evaluated using silhouette score, total distance to medoids, and minimum cluster share. The selected solution uses four balanced clusters.

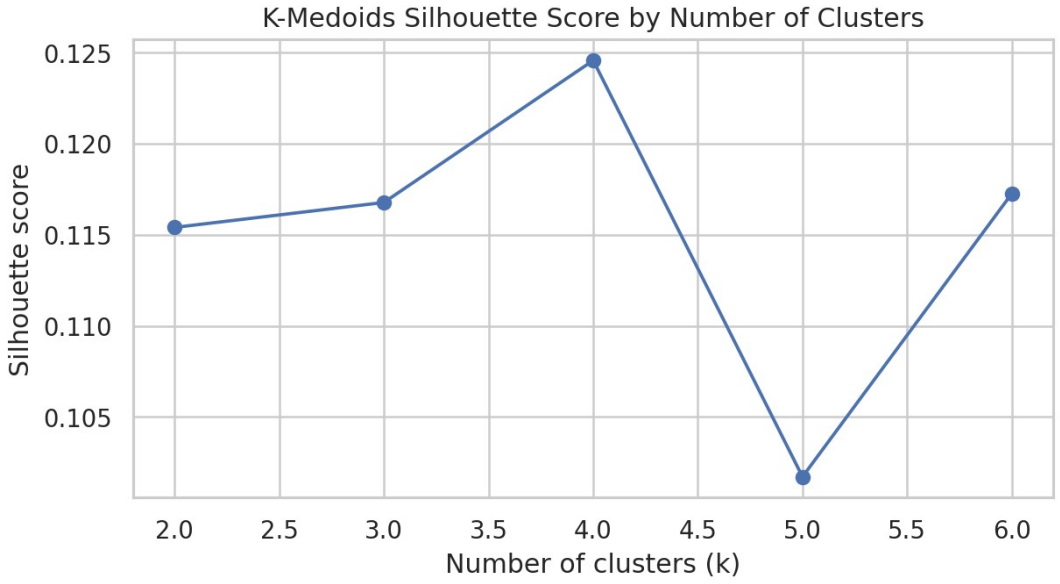


Figure 5. K-Medoids silhouette comparison by k.

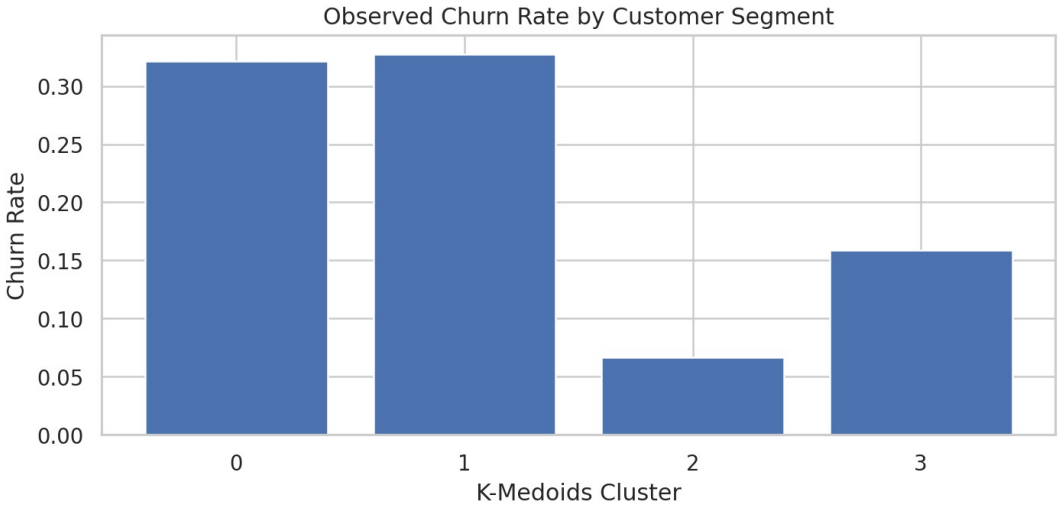


Figure 6. Observed churn rate by customer segment.

Cluster profiles

Cluster	Customers	Churn	Avg Age	Avg Balance	Active Rate
0	2,254	32.1%	40.7	95,602	32.9%
1	2,078	32.7%	49.6	96,410	36.7%
2	2,897	6.7%	34.8	21,941	65.7%
3	2,771	15.9%	33.8	103,020	63.0%

5. Hierarchical Clustering Benchmark

Single, complete, and average linkage are benchmarked across k=2..6. A key improvement in this portfolio edition is that silhouette score is not used blindly: configurations with a cluster smaller than 5% of the sample are flagged as operationally unbalanced. This prevents one-point outlier clusters from being presented as strong segmentation solutions.

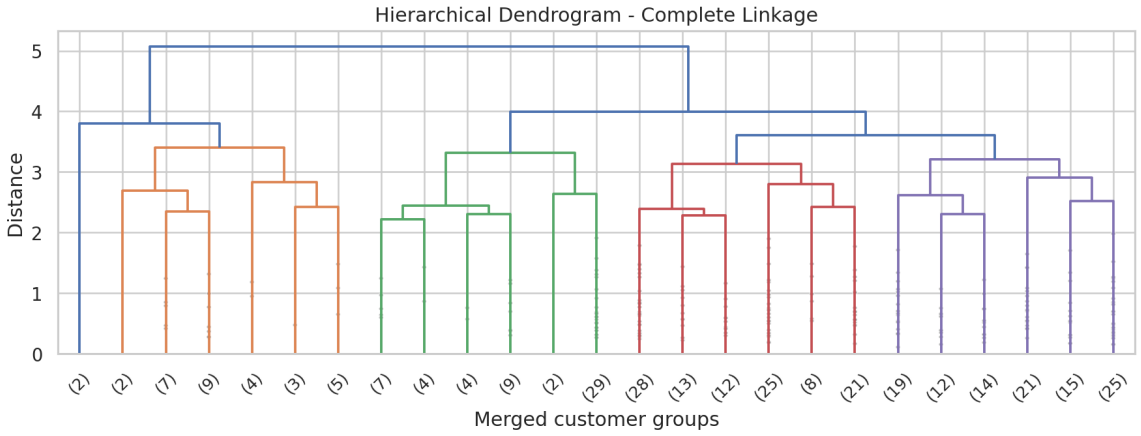


Figure 7. Complete-linkage dendrogram on a readable customer sample.

Model comparison

Model	Configuration	Silhouette	Role
K-Medoids	k=4	0.125	Primary customer segmentation
Hierarchical	complete, k=2	0.199	Benchmark / validation

The hierarchical benchmark achieves a higher silhouette score with two broad groups, while K-Medoids is retained as the primary segmentation because it provides four balanced and more granular business segments. This is a deliberate trade-off between separation and actionable segmentation detail.

6. Explainable Retention Prioritization

The fuzzy decision layer converts customer balance, activity status, and a relative cluster-risk index into a 0-100 retention-priority score. The relative risk index is min-max normalized across observed cluster churn rates, avoiding the original problem where “high-risk” fuzzy memberships started above the actual risk range.

Decision inputs

- Balance: low / medium / high using data-driven balance quantiles.
- Activity: inactive / active.
- Cluster risk index: low / medium / high on a relative 0-100 scale.
- Output: Low, Moderate, or High retention priority.

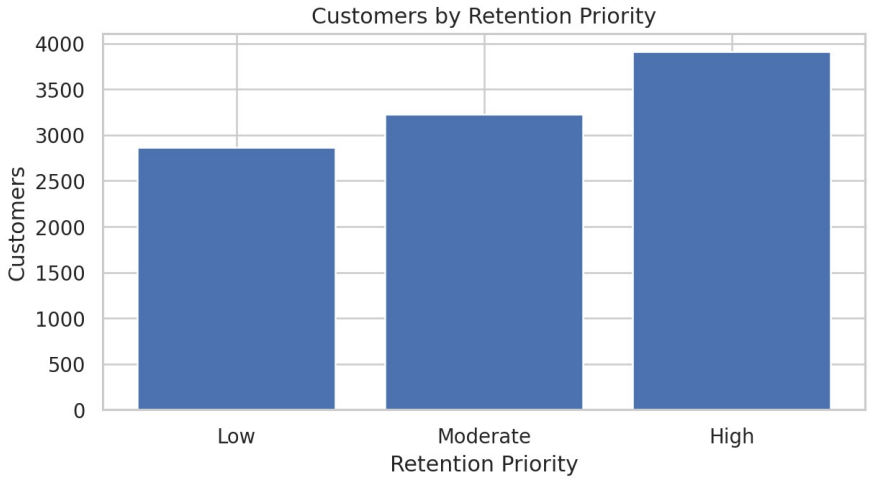


Figure 8. Customer counts by retention priority.

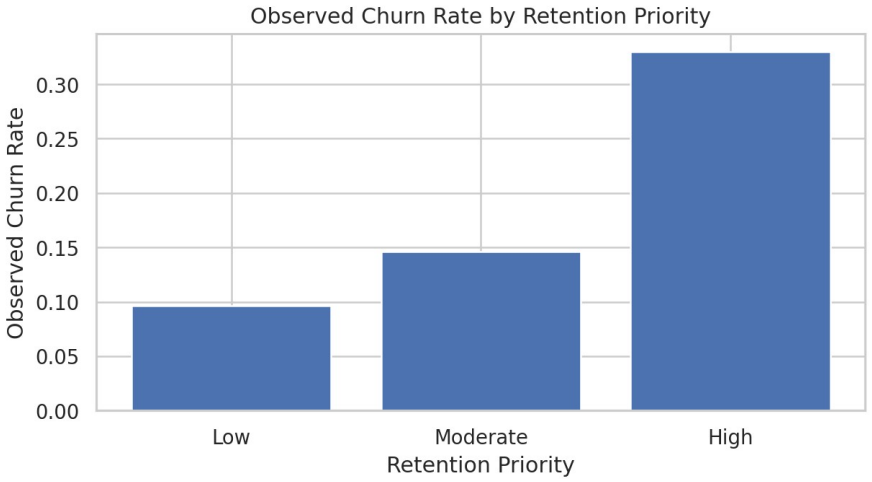


Figure 9. Higher priority groups show higher observed churn, supporting the ranking logic.

Priority summary

Priority	Customers	Share	Observed Churn
Low	2,861	28.6%	9.6%
Moderate	3,230	32.3%	14.6%
High	3,909	39.1%	33.0%

7. Business Recommendations & Conclusion

Recommendations

- Prioritize re-engagement campaigns for inactive customers, where observed churn is materially higher.
- Use segment-specific retention offers rather than applying the same incentive to all customers.
- Review high-risk, higher-balance segments before allocating expensive incentives because potential value loss may be larger.
- Use the fuzzy score as a transparent prioritization queue that can be inspected through fired rules.
- Run A/B tests on retention actions and measure incremental churn reduction; descriptive associations do not establish causality.

Technical strengths

- Fully reproducible random seeds.
- Consistent preprocessing for training data and new customer records.
- No automatic package installation inside the notebook.
- No dependency on scikit-learn-extra or scikit-fuzzy.
- Portfolio-ready exports for KPIs, cluster profiles, model comparison, priorities, and charts.
- Notebook executes from top to bottom with no saved errors.

Conclusion

This project demonstrates a complete analytics workflow from raw customer data to business-facing segmentation and prioritization. The strongest insight is the gap between active and inactive customers, while the four K-Medoids segments provide additional context for targeted retention. The final priority layer translates the analytical findings into an explainable action queue suitable for portfolio demonstration and further business testing.

Project type: Individual portfolio project - developed independently.